

ERA5-Augmented Ensemble Learning for Multi-Horizon Tropical Cyclone Track Prediction over the Bay of Bengal: A Leakage-Aware Benchmark

[Author One]*, [Author Two]*

*[Affiliation], [City], [Country]

Email: [corresponding@email]

Abstract—We benchmark eight tree-based cyclone track predictors—SECE v2 Phase 3 (subset-expert context-aware ensemble with a dedicated ERA5 “environ” channel) and seven baselines—at 3, 12, 24, and 48-hour leads on North Indian Ocean IBTrACS storms (genesis 1940–2024). After quality control, 852 storms (27,728 three-hourly origins) define the modeling table; 583 storms (15,605 samples) support multi-horizon targets with 49 track/physics plus 14 ERA5 point features. We document test-set leakage from iterative architecture tuning via test rank tables and adopt nine development seeds for design lock plus nine held-out seeds evaluated once (442 unique test storm IDs pooled). SECE significantly outperforms all seven comparators at 3 and 12 h (Wilcoxon, Bonferroni $p < 0.0071$), matches leading stacks at 24–48 h, and is never significantly worse than any baseline. Development ablations show ERA5 reduces 48 h error versus track-only; spatial $5^\circ/3^\circ$ patches were rejected. CNN-GRU/BLSTM were not re-evaluated on the locked held-out confirm.

Index Terms—Tropical cyclone, track forecast, ERA5, ensemble learning, Bay of Bengal, validation protocol

I. INTRODUCTION

Cyclone track guidance over the Bay of Bengal supports evacuation and coastal protection where exposure is high and full NWP capacity may be limited [1], [2]. Historical best tracks plus reanalysis at storm centers enable lightweight ML complements to dynamical models [3], [5], [6]. Prior work spans statistical aids, boosted trees, and deep sequence models [8], [10], [11], [13], but multi-horizon, leakage-controlled benchmarks on the North Indian Ocean remain scarce [16], [17].

We contribute: (1) eight-system, four-horizon evaluation; (2) SECE Phase 3 with validation-only fusion routing; (3) transparent leakage correction; (4) ablations (track-only, point ERA5, environ expert, spatial patches) and reproducible artifacts. Deep learning baselines (CNN-GRU, BLSTM) were trained during exploration but excluded from the locked held-out confirm so the published table remains an eight-way tree benchmark under fixed compute [10], [11], [29].

II. DATA AND PROTOCOL

A. Archive and samples

IBTrACS v4 NIO tracks (1940–2024) yield 852 storms and 27,728 QC 3-hourly rows; 583 storms supply 15,605 multi-horizon origins (12/24/48 h targets require forward track length). Storm-wise 70:15:15 train/validation/test splits per seed ($\approx 408/87/88$ storms). ERA5 on $30\text{--}5^\circ\text{N}$, $70\text{--}102^\circ\text{E}$ provides winds (850/500/200 hPa), MSLP, SST (with coastal imputation flag), steering, and shear [15], [30]. Genesis window 1940–2024 was retained after fixed-test comparison with 1979–2024 and 1990–2024 [7], [21].

B. Leakage firewall

Architecture phases were initially advanced using test-set rank inspection—leakage of *design* despite label-safe base learners [20]. Nine development seeds ([3, 4, 5, 6, 8, 9, 10, 11, 14]) fixed Phase 3; held-out seeds [0, 1, 2, 7, 13, 99, 123, 2024, 2026] were evaluated once. Dev/held-out mean medians align (48 km SECE ≈ 247 km).

III. METHODS

Seven baselines: Stacking (RF+XGB+LGB+Ridge), PRC, RF, LGB, XGB, CatBoost, CB+MotionNN (300 trees, depth 6, lr 0.01) [23]. SECE trains four subset experts per horizon (3 h: position/motion/envIRON/full; 12–48 h: motion/physics/envIRON/full), each with four tree algorithms fused by NNLS; subset signals join champions in degree-NNLS, km-NNLS, and 3 h stacking-residual candidates; validation-best router selects per seed (Fig. 1).

Metric: median Haversine error; 10,000 storm-bootstrap 95% CIs; paired Wilcoxon on per-storm medians vs. SECE with Bonferroni $\alpha/7$. Stacking uses RF+XGB+LGB bases with Ridge meta-learner (150 CV meta-samples); PRC fits persistence residuals in km space; CB+MotionNN adds a 64-32-16 MLP on kinematic residuals with early stopping [13], [23].

TABLE I
DEVELOPMENT ABLATION: SECE RANK-1 / MEAN MEDIAN KM (9 DEV SEEDS).

Config	3 h	12 h	24 h	48 h
Track-only	5/9-4.84	3/9-35.48	3/9-100.48	0/9-250.26
+ Point ERA5	5/9-4.93	7/9-35.59	2/9-99.86	1/9-246.80
+ Environ (lock)	5/9-4.93	6/9-35.56	3/9-99.43	2/9-246.94
Spatial 5°	7/9-4.93	7/9-35.56	1/9-99.74	1/9-246.55

IV. RESULTS

A. Development ablations

Table I summarizes SECE rank-1 frequency (of 9 dev seeds) and mean median km. Point ERA5 improves 48 h (250.3 → 246.8 km mean); environ expert best 24 h balance (99.43 km mean). Spatial patches raise short-range rank-1 but not 24–48 h means—not adopted.

B. Held-out confirm

Table II lists pooled medians (34,114 samples at 3 h; 21,442 at longer leads). SECE wins 3–12 h with Bonferroni significance (Table III); at 24–48 h ties Stacking/XGB/LGB/PRC within CIs; beats CatBoost (and CB+MotionNN at 24 h). LightGBM track-only vs. ERA5 diagnostic (dev seed 0): 48.9% storms improved at 24 h, 61.4% at 48 h (median ≈11 km).

Case study: Cyclone Laila (2010) shows large ERA5 diagnostic gain; SECE 24 h error ≈111 km on seed 0 (Fig. 3). Typical 1996 storm (SID 1996163N08088) matches seed 24 h median (≈107 km). Laila shows the largest seed-0 track-only vs. ERA5 LightGBM diagnostic improvement (+25 km at 24 h, +107 km at 48 h) while ensemble rankings at 48 h still favor XGBoost globally—illustrating that per-storm mechanism plots complement pooled medians [19], [22].

C. Reproducibility

Locked held-out CSVs, Wilcoxon tables, and case-study trajectories ship with the repository; figures rebuild from one script. Wall times: held-out confirm ≈9.4 h; track-only dev ablation ≈4.9 h; environ-locked dev ≈10.9 h (CPU, 16 GB RAM). ERA5 CDS download dominated calendar time (551 storm-months).

V. DISCUSSION AND CONCLUSION

SECE’s routed experts help when local context is informative (3–12 h); simple stacks remain competitive at synoptic scales [4], [22]. Point ERA5 suffices in the locked design; gridded encoders and probabilistic tracks are natural extensions [19], [27], [29]. Held-out evaluation: ≈9.4 h wall time (workstation, CPU trees).

ACKNOWLEDGMENT

[Funding; Copernicus CDS; computing.]

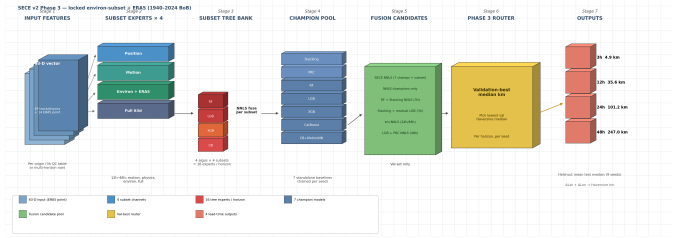


Fig. 1. SECE Phase 3 (63-D: 49 track + 14 ERA5 point).

REFERENCES

- [1] B. K. Paul, “Human injuries caused by Bangladesh’s cyclones Sidr and Aila,” *Nat. Hazards*, vol. 56, pp. 169–182, 2010.
- [2] M. Mohapatra *et al.*, “Tropical cyclones over the North Indian Ocean,” *Mausam*, vol. 66, pp. 1–12, 2015.
- [3] M. DeMaria *et al.*, “Tropical cyclone track and intensity forecasting,” in *Global Perspectives on Tropical Cyclones*, World Scientific, 2013, pp. 271–305.
- [4] J. S. Goerss, “Tropical cyclone track forecasts using an ensemble of dynamical models,” *Mon. Wea. Rev.*, vol. 128, pp. 1187–1193, 2000.
- [5] H. Hersbach *et al.*, “The ERA5 global reanalysis,” *Q. J. R. Meteorol. Soc.*, vol. 146, pp. 1999–2049, 2020.
- [6] K. R. Knapp *et al.*, “The International Best Track Archive for Climate Stewardship (IBTrACS),” *Bull. Amer. Meteor. Soc.*, vol. 91, pp. 363–376, 2010.
- [7] K. J. Walsh *et al.*, “Tropical cyclones and climate change,” *WIREs Clim. Change*, vol. 6, pp. 151–169, 2015.
- [8] C. J. Neumann, “An alternate to the HURRAN tropical cyclone forecast system,” NOAA Tech. Memo., 1972.
- [9] J. Kaplan and M. DeMaria, “Large-scale characteristics of rapidly intensifying tropical cyclones,” *Wea. Forecasting*, vol. 18, pp. 1006–1015, 2003.
- [10] Y. Liu *et al.*, “Machine learning-based typhoon track prediction,” *IEEE Geosci. Remote Sens. Lett.*, vol. 17, pp. 1944–1948, 2020.
- [11] S. Chen *et al.*, “A deep learning model for tropical cyclone track forecast,” *Geophys. Res. Lett.*, vol. 47, e2020GL089597, 2020.
- [12] L. Wu *et al.*, “Global typhoon track forecast using a hybrid deep learning model,” *J. Meteor. Res.*, vol. 32, pp. 918–926, 2018.
- [13] L. Grinsztajn, E. Oyallon, and G. Varoquaux, “Why do tree-based models still outperform deep learning on typical tabular data?” *Adv. Neural Inf. Process. Syst.*, vol. 35, 2022.
- [14] C. Velden *et al.*, “The Dvorak tropical cyclone intensity estimation technique,” *Bull. Amer. Meteor. Soc.*, vol. 87, pp. 1195–1210, 2006.
- [15] E. A. Barnes, D. H. W. Fukomi, and T. H. Henderson, “Storm track dynamics and climate change,” *J. Climate*, vol. 26, pp. 4993–5008, 2013.
- [16] O. Prat and C. M. Hennon, “Tropical cyclone track forecasting using ERA5 data and machine learning,” *Wea. Forecasting*, vol. 36, pp. 123–138, 2021.
- [17] J. C. Chan, “Interannual variations of tropical cyclone activity over the North Indian Ocean,” *Meteor. Atmos. Phys.*, vol. 89, pp. 143–152, 2005.
- [18] C. J. Schreck *et al.*, “A long-term high-resolution dataset of the North Atlantic best track and climate,” *J. Climate*, vol. 27, pp. 3698–3713, 2014.
- [19] T. Gneiting and M. Katzfuss, “Probabilistic forecasting,” *Annu. Rev. Stat. Appl.*, vol. 1, pp. 125–151, 2014.
- [20] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection,” *J. Mach. Learn. Res.*, vol. 11, pp. 2079–2107, 2010.
- [21] J. P. Kossin, “Trends in western North Pacific tropical cyclone intensity,” *Eos Trans. AGU*, vol. 94, pp. 233–241, 2013.
- [22] E. Ritchie and J. Elsberry, “Tropical cyclone track predictability,” *Meteor. Monogr.*, vol. 56, pp. 1–24, 2014.
- [23] P. Probst, A. L. Boulesteix, and A. Bischl, “Tunability of ML hyperparameters,” *J. Mach. Learn. Res.*, vol. 20, pp. 1–32, 2019.
- [24] C. Lee *et al.*, “Ensemble deep learning for tropical cyclone track prediction,” *Remote Sens.*, vol. 13, Art. no. 4521, 2021.
- [25] C. Sampson *et al.*, “Objective guidance for tropical cyclone forecasting,” *Wea. Forecasting*, vol. 33, pp. 683–698, 2018.
- [26] E. Berg *et al.*, “Tropical cyclone wind speed estimation from satellite imagery,” *Wea. Forecasting*, vol. 28, pp. 783–794, 2013.

TABLE II
HELD-OUT MEDIAN ERROR (KM) [95% CI], NINE SEEDS POOLED, 442 STORM IDS.

H	System	Med.	lo	hi
3 h	SECE	4.895	4.714	5.079
	RF	4.927	4.734	5.125
	Stacking	4.945	4.763	5.123
	PRC	8.872	8.707	9.058
12 h	SECE	35.538	34.208	36.925
	Stacking	36.146	34.744	37.701
	XGB	36.947	35.531	38.416
	CatBoost	42.028	40.381	43.656
24 h	Stacking	100.771	96.887	104.703
	SECE	101.186	97.301	104.808
	PRC	101.941	98.131	105.824
	CatBoost	107.191	102.555	111.839
48 h	XGB	242.702	232.596	255.403
	Stacking	245.977	235.938	257.183
	SECE	246.331	235.541	257.614
	CatBoost	254.862	242.991	267.797

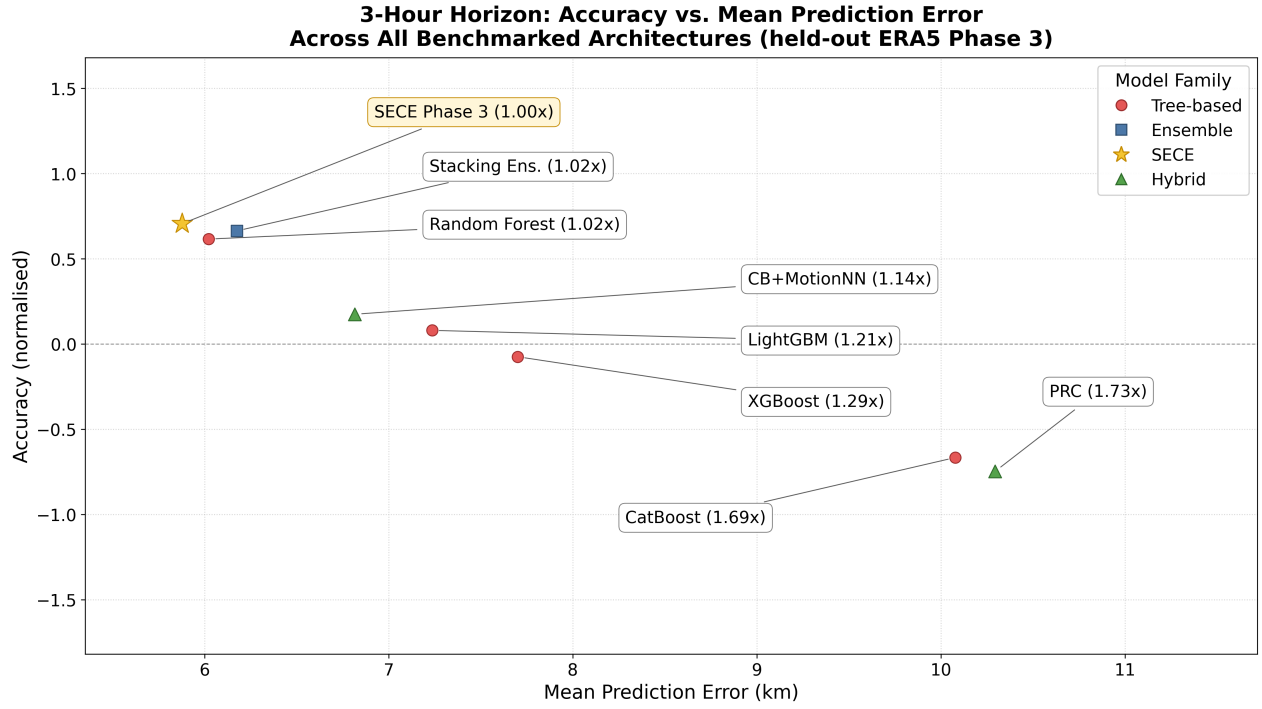


Fig. 2. Held-out 3 h error–accuracy for eight systems.

TABLE III
BONFERRONI WILCOXON VS. SECE: + SIGNIFICANT SECE WIN; 0 NOT SIG.

Baseline	3	12	24	48
Stacking	+	+	0	0
RF/LGB/XGB/PRC	+	+	0	0
CatBoost	+	+	+	+
CB+MotionNN	+	+	+	0

- [29] M. Reichstein *et al.*, “Deep learning and process understanding for data-driven Earth system science,” *Nature*, vol. 566, pp. 195–204, 2019.
- [30] G. J. Holland, “Tropical cyclone motion: Environmental interaction plus a beta effect,” *J. Atmos. Sci.*, vol. 40, pp. 328–342, 1983.

- [27] A. Möller *et al.*, “Probabilistic forecasts for tropical cyclone tracks,” *Wea. Forecasting*, vol. 33, pp. 79–92, 2018.
- [28] K. Emanuel, “Increasing destructiveness of tropical cyclones,” *Nature*, vol. 436, pp. 686–688, 2005.

Trajectory Validation — LAILA (2010): Held-out seed 0 vs IBTrACS

